

Leon Meng

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EDUCATION

University of California, Irvine

Bachelor of Computer Science and Engineering

Irvine, CA
2025 - 2029

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML, CSS

Web & Backend: React, Vite, Tailwind, FastAPI, Supabase, PostgreSQL, REST APIs, Vercel serverless functions

Robotics & Vision: Odometry & localization, path planning (Roadrunner), finite state machines, OpenCV, Limelight 3A vision, CAD (Fusion 360), photogrammetry (RealityScan), 3D printing

Data & Tooling: pandas, NumPy, SQLite, Streamlit, Git, Linux

EXPERIENCE

SomaFuture

Software Engineer/Consultant

Oeiras, Portugal
Jun 2026 - Aug 2026

- Built a functional Next.js/Supabase prototype of the company's B2B materials platform to define product scope and technical requirements, directly informing a formal vendor RFP for the platform's outsourced development
- Built a Python document-translation pipeline that OCRs scanned Portuguese technical PDFs, machine-translates the extracted text to Simplified Chinese, and reassembles it into the original layout with embedded CJK fonts — eliminating manual reformatting for the team
- Evaluated three approaches for digitizing the product catalog into 3D — photogrammetry scanning (RealityScan), CAD-authored modeling, and an AI photo-to-3D tool — weighing render quality against BIM-readiness to define the platform's 3D-asset requirements
- Led research, outreach, and contract negotiation with vendor finalists — BiMaYun (BIM/3D software) and Stardecerp (website and backend development) — coordinating with the company's product, BIM engineering, and legal teams through final contract signing

First Tech Challenge — Teams 16031 & 16205

Autonomous & Mechanical Engineer

Vancouver, BC
Oct 2021 - Jun 2025

- Implemented an odometry system using encoders; computed horizontal/vertical displacement per time step, corrected for rotational drift, and converted to field coordinates via trigonometric transforms for sub-centimeter localization accuracy
- Built a vision-guided object detection system using Limelight 3A and OpenCV: designed a color-isolation pipeline (contour detection, color filtering) to detect and localize game elements in real time, then computed target contour angle and position to autonomously align the robot for pickup
- Implemented a finite state machine coordinating all robot mechanisms via event triggers, and integrated Roadrunner for trajectory planning and tuning to produce repeatable multi-step autonomous routines
- Designed and fabricated custom 3D-printed mechanical components (chassis, drivetrain, arm pivot, scoring mechanism) and co-authored the team's Engineering Portfolio; earned multiple top-scoring awards at the British Columbia and Alberta championships

PROJECTS

GARDEROBE

Full-Stack Wardrobe Manager — React, Supabase, FastAPI, Claude API

the-garderoberoom.com

- Built and shipped to production a full-stack wardrobe management app with AI auto-tagging that extracts item name, brand, color, and type from an uploaded photo via the Anthropic Claude API
- Built a walkable 3D gallery to showcase the wardrobe using only CSS, with hand-built perspective, textures, and performance tuning for smooth navigation
- Built a full social layer on Supabase realtime — profiles, publishing, an Explore page, a people directory, friends, and likes — plus a feed aggregating fashion and culture articles from multiple third-party news APIs

DoMusMat

Full-Stack B2B Materials Platform — Next.js, Supabase, PostgreSQL, WebGL

do-mus-mat-v2.vercel.app

- Built the full-stack web app end-to-end with Next.js and Supabase, standing up a PostgreSQL schema and API layer to drive dynamic product pages, syncing the live catalog across lighting, flooring, doors, and other categories from the company's existing WordPress site
- Built a WebGL-based 3D room configurator for real-time in-browser product placement, with a "render in HD" step that ships the configured scene to Modal serverless GPUs for photorealistic rendering via Blender Cycles

EquityEval — Equity Valuation Workbench

Financial Research System — Python, FastAPI, PostgreSQL/TimescaleDB, Redis, Next.js, TypeScript

github.com/ggttlplp201/EquityEval

- Built a local-first equity research platform that ingests SEC XBRL filings and market data into point-in-time financial histories, keeping a full trail of restatements, sources, and data quality for every value
- Built a Python valuation engine that works backward from a company's market price to the operating performance it implies, with versioned assumptions, scenario/sensitivity analysis, and 3/5/10-year historical comparisons
- Built a Next.js research workbench with company analysis, filing monitoring, and a graph-based Sector Explorer, backed by PostgreSQL-orchestrated workflows and a generated FastAPI layer

AWARDS & ACHIEVEMENTS

- USACO Gold Division (2023)